

ECO 310: Complete Guide to WARP Questions on Midterms

All Question Types and How to Solve Them

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1 Overview

WARP (Weak Axiom of Revealed Preference) appears on almost every ECO 310 midterm, but in different formats. This guide covers ALL the types you might see.

1.1 What is WARP?

WARP says: If you choose bundle A when bundle B was also affordable, you cannot later choose bundle B when bundle A is also affordable.

Why? The first choice revealed you prefer A to B. The second choice reveals you prefer B to A. This is a contradiction!

1.2 The Two Key Conditions for WARP Violation

For a new bundle (x_1, x_2) to violate WARP when the original bundle was (x_1^*, x_2^*) :

1. **Condition 1 (Past):** The new bundle was affordable at old prices

$$p_1^{old}x_1 + p_2^{old}x_2 \leq m^{old}$$

2. **Condition 2 (Present):** The old bundle is affordable at new prices (proven by new bundle costing enough)

$$p_1^{new}x_1 + p_2^{new}x_2 \geq p_1^{new}x_1^* + p_2^{new}x_2^*$$

If BOTH hold $\Rightarrow$ WARP VIOLATION!

2 Type 1: Given Costs, Determine if WARP is Violated

2.1 Format

You're told:

- What someone bought on Day 1 and how much it cost
- What they bought on Day 2 and how much it cost
- How much one bundle would have cost at the other day's prices

Question: Does this violate WARP?

2.2 Example: Spring 2023 Problem 2b

Problem

Previously, he consumed $(4, 2)$ when prices were $(1, 3)$. Prices are now $(3, 5)$. Which bundles (x_1, x_2) would violate WARP if purchased now?

Solution Strategy:

Step 1: Calculate the old budget

$$m^{old} = 1(4) + 3(2) = 10$$

Step 2: Calculate what the old bundle costs at new prices

$$\text{Cost}_{(4,2)}^{new} = 3(4) + 5(2) = 22$$

Step 3: Write the two inequalities

$$\text{Inequality 1: } x_1 + 3x_2 \leq 10 \quad (\text{was affordable})$$

$$\text{Inequality 2: } 3x_1 + 5x_2 \geq 22 \quad (\text{costs enough now})$$

Step 4: Any bundle satisfying BOTH inequalities violates WARP!

Examples:

- $(6, 1)$: Old cost $= 9 \leq 10$ *eckmark*, New cost $= 23 \geq 22$ *eckmark* $\Rightarrow$ **VIOLATION**
- $(3, 2)$: Old cost $= 9 \leq 10$ *eckmark*, New cost $= 19 \not\geq 22$ *imes* $\Rightarrow$ **SAFE**

3 Type 2: Given Cross-Price Information (Multiple Choice)

3.1 Format

You're told:

- What bundle A cost on Monday and what it cost on Tuesday
- What bundle B cost on Monday and what it cost on Tuesday
- What choice was made each day

Question: Under what condition is WARP violated?

3.2 Example: Spring 2024 Q9

Problem

On Monday, Homer buys a bundle for \$100. On Tuesday, he buys a different bundle for \$80, which would have cost \$100 on Monday. In which situation does Homer violate WARP?

- (a) If Monday bundle would cost more than \$80 at Tuesday prices
- (b) If Monday bundle would cost less than \$80 at Tuesday prices
- (c) If Monday bundle would cost less than \$100 at Tuesday prices
- (d) Homer does not violate WARP
- (e) Not enough information

Solution Strategy:

Step 1: Identify what we know

- Monday: Bought bundle M for \$100
- Tuesday: Bought bundle T for \$80
- Bundle T would have cost \$100 on Monday

Step 2: Check the two conditions

Condition 1: Was bundle T affordable on Monday?

- Bundle T cost \$100 on Monday
- Homer spent \$100 on Monday (his budget)
- So YES, bundle T was affordable on Monday *eckmark*

Condition 2: Was bundle M affordable on Tuesday?

- Homer bought bundle T for \$80 on Tuesday
- This proves his budget was at least \$80
- Bundle M is affordable if it costs $\leq$ \$80 at Tuesday prices

Step 3: Determine violation condition

For WARP violation, we need BOTH conditions:

1. Bundle T was affordable Monday *eckmark* (already true)
2. Bundle M is affordable Tuesday $\Leftrightarrow$ costs $\leq$ \$80 at Tuesday prices

Answer: (b) If Monday bundle would cost less than \$80 at Tuesday prices

3.3 Why This Works

Monday: He chose M over T when both cost \$100 (both affordable) $\Rightarrow M \succ T$

Tuesday: If M costs $< \$80$, then both M and T are affordable (he has $\geq \$80$), but he chose T
 $\Rightarrow T \succ M$

Contradiction!

4 Type 3: Spring 2025 Version (Slight Variation)

Problem

On Monday, Homer buys a bundle for \$90. On Tuesday, he buys a different bundle for \$100, that would have cost \$80 at Monday prices. In which situation does he violate WARP?

- (a) If Monday bundle would cost \$100 at Tuesday prices
- (b) If Monday bundle would cost more than \$100 at Tuesday prices
- (c) Homer does not violate WARP

Solution:

Given:

- Monday: Bought bundle M for \$90 (so budget = \$90)
- Tuesday: Bought bundle T for \$100 (so budget = \$100)
- Bundle T would have cost \$80 on Monday

Check Condition 1: Was T affordable Monday?

- T cost \$80 on Monday
- Budget was \$90
- $\$80 \leq \90 *eckmark* YES, T was affordable!

Check Condition 2: Was M affordable Tuesday?

- Homer spent \$100 Tuesday (his budget)
- M is affordable if it costs $\leq \$100$ at Tuesday prices

For WARP violation:

- Condition 1: *eckmark* (T was affordable Monday)
- Condition 2: M costs $\leq \$100$ at Tuesday prices

Answer: (a) If Monday bundle would cost \$100 at Tuesday prices (or less!)

Actually both (a) and anything $\leq \$100$ would work, but (b) says "more than \$100" which would NOT violate WARP because then M wouldn't be affordable Tuesday.

5 Step-by-Step Solution Template

Use this template for ANY WARP problem:

5.1 Template

WARP Solution Template

Given Information:

- Time 1: Bought bundle (x_1^*, x_2^*) with prices (p_1^{old}, p_2^{old})
- Budget/Cost at Time 1: m^{old}
- Time 2: Considering bundle (x_1, x_2) with prices (p_1^{new}, p_2^{new})

Step 1: Calculate old bundle cost at new prices

$$C_{old}^{new} = p_1^{new} x_1^* + p_2^{new} x_2^*$$

Step 2: Write Condition 1 (Was new bundle affordable before?)

$$p_1^{old} x_1 + p_2^{old} x_2 \leq m^{old}$$

Step 3: Write Condition 2 (Is old bundle affordable now?)

$$p_1^{new} x_1 + p_2^{new} x_2 \geq C_{old}^{new}$$

Step 4: Check both conditions

- If BOTH true $\Rightarrow$ **WARP VIOLATION**
- If either false $\Rightarrow$ **NO VIOLATION (safe)**

6 Common Tricks and Pitfalls

6.1 Trick 1: "Would have cost" statements

When a problem says "bundle X would have cost \$Y at time 2 prices," this is giving you DIRECT information about the cost. Use it!

Example: "Tuesday bundle would have cost \$100 at Monday prices"

$$\Rightarrow p_1^{Mon} x_1^{Tue} + p_2^{Mon} x_2^{Tue} = 100$$

6.2 Trick 2: Budget vs. Cost

- If someone **buys** a bundle for \$X, their budget is **at least** \$X
- If we know they spent exactly their whole budget (common assumption), budget = cost

6.3 Trick 3: Inequality direction

ALWAYS:

- Past affordability: $\leq$ (bundle was cheap enough)
- Present affordability: $\geq$ (bundle costs enough to prove old one affordable)

Never flip these!

6.4 Trick 4: "Exactly at the boundary"

If a bundle costs **exactly** the budget amount (equality), it's still affordable!

- $\$100 \leq \100 *eckmark* TRUE
- This is a boundary case

7 Practice Problems

7.1 Problem 1

Last week: Bought $(3, 5)$ at prices $(2, 4)$

This week: Prices are $(5, 3)$

Which bundles violate WARP if purchased this week?

- (a) $(6, 2)$
- (b) $(2, 5)$
- (c) $(4, 4)$

7.2 Problem 2

Monday: Homer buys bundle M for \$50

Tuesday: Homer buys bundle T for \$60, which would have cost \$40 on Monday

WARP is violated if:

- (a) Bundle M costs less than \$60 at Tuesday prices
- (b) Bundle M costs more than \$60 at Tuesday prices
- (c) Never violated

7.3 Problem 3

Day 1: Prices $(1, 2)$, bought $(10, 5)$

Day 2: Prices $(3, 1)$, bought $(4, 8)$

Does this violate WARP? Show your work.

8 Solutions to Practice Problems

8.1 Solution 1

Given:

- Old: $(3, 5)$ at prices $(2, 4)$, cost $= 2(3) + 4(5) = 26$
- New prices: $(5, 3)$
- Old bundle now costs: $5(3) + 3(5) = 30$

Conditions:

$$\text{Condition 1: } 2x_1 + 4x_2 \leq 26$$

$$\text{Condition 2: } 5x_1 + 3x_2 \geq 30$$

Check each:

(a) $(6, 2)$:

- Old cost: $2(6) + 4(2) = 20 \leq 26$ *eckmark*
- New cost: $5(6) + 3(2) = 36 \geq 30$ *eckmark*
- **VIOLATES WARP!**

(b) $(2, 5)$:

- Old cost: $2(2) + 4(5) = 24 \leq 26$ *eckmark*
- New cost: $5(2) + 3(5) = 25 \not\geq 30$ *imes*
- **SAFE**

(c) $(4, 4)$:

- Old cost: $2(4) + 4(4) = 24 \leq 26$ *eckmark*
- New cost: $5(4) + 3(4) = 32 \geq 30$ *eckmark*
- **VIOLATES WARP!**

8.2 Solution 2

Given:

- Monday: M costs \$50 (budget = \$50)
- Tuesday: T costs \$60 (budget = \$60)
- T would have cost \$40 on Monday

Condition 1: Was T affordable Monday?

- T cost \$40, budget was \$50
- $\$40 \leq \50 *eckmark*

Condition 2: Is M affordable Tuesday?

- Budget Tuesday is \$60
- M is affordable if it costs $\leq$ \$60

Answer: (a) If M costs less than \$60 at Tuesday prices (or equal!)

8.3 Solution 3

Day 1:

- Bought: $(10, 5)$
- Prices: $(1, 2)$
- Cost: $1(10) + 2(5) = 20$

Day 2:

- Bought: $(4, 8)$
- Prices: $(3, 1)$
- Cost: $3(4) + 1(8) = 20$

Check if Day 2 bundle was affordable Day 1:

$$1(4) + 2(8) = 20 \leq 20 \quad \checkmark$$

Check if Day 1 bundle is affordable Day 2:

$$3(10) + 1(5) = 35$$

Day 2 budget is \$20 (what they spent). Is $\$35 \leq \20 ? NO!

Conclusion: NO VIOLATION - Day 1 bundle is NOT affordable on Day 2.